propagating through space is actually accompanied by a surrounding cloud of virtual particles.



As a result, it's helpful to distinguish between the "bare" fields as they would behave in an imaginary world where all interactions were simply turned off, and the "physical" fields that are accompanied by other fields they interact with. The infinities that you get by naïvely turning a crank in the Feynman diagrams are simply a result of trying to work with bare fields, whereas what we really observe are physical ones. The adjustment required to go from one to another is sometimes informally described as "subtracting off infinity to get a finite answer," but that's misleading. No physical quantities are infinite, nor were they ever; the infinities that quantum-field-theory pioneers managed to "hide" were simply artifacts of the very big difference between fields that interact and fields that don't. (We face exactly this kind of issue when trying to estimate the vacuum energy in quantum field theory.)

Nevertheless, renormalization comes with important physical insights. When we want to measure some property of a particle, such as its mass or charge, we probe it by seeing how it interacts with other particles. Quantum field theory teaches us that the particles we see aren't simple point-like objects; each particle is surrounded by a cloud of other virtual particles, or (more accurately) by the other quantum fields it interacts with. And interacting with a cloud is different from interacting with a point. Two particles that smash into each other at high velocity will penetrate deep into each other's clouds, seeing relatively compact vibrations, while two particles that pass by slowly will see each other as (relatively) big puffy balls. Consequently, the apparent mass or charge of a particle will depend on the energy of the probes with which we look at it. This isn't just a song and dance: it's an experimental prediction, which has been seen unmistakably in particle-physics data.